

Неравенство Прекопы–Лейндлера

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The functional form of the Brunn–Minkowski inequality. It is proved by induction on the dimension starting from the one-dimensional case, and then gives almost for free the log-concavity of marginals, of convolutions and of the Gaussian measure.

Definition 1. $f: \mathbb{R}^n \rightarrow [0, +\infty)$ is **log-concave** if $f(\lambda x + (1 - \lambda)y) \geq f(x)^\lambda f(y)^{1-\lambda}$ for all x, y and $\lambda \in (0, 1)$; equivalently $f = e^{-V}$ with V convex.

Definition 2. $\lambda A + (1 - \lambda)B = \{\lambda a + (1 - \lambda)b : a \in A, b \in B\}$; $|\cdot|$ is Lebesgue measure.

Theorem 1 (Prékopa 1971, Leindler 1972). Let $\lambda \in (0, 1)$ and let $f, g, h: \mathbb{R}^n \rightarrow [0, +\infty)$ be measurable with $h(\lambda x + (1 - \lambda)y) \geq f(x)^\lambda g(y)^{1-\lambda}$ for all x, y . Then $\int h \geq (\int f)^\lambda (\int g)^{1-\lambda}$.

Theorem 2 (Brunn–Minkowski). $|A+B|^{1/n} \geq |A|^{1/n} + |B|^{1/n}$ for nonempty compact $A, B \subseteq \mathbb{R}^n$.

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1. Let $A, B \subseteq \mathbb{R}$ be nonempty compact sets. Prove that $|A + B| \geq |A| + |B|$. (*Hint: translate so that $\max A = \min B = 0$.*)
 2. Prove Theorem 1 for $n = 1$. (*Hint: one may assume $\sup f = \sup g = 1$; for $0 < t < 1$ check that $\{h \geq t\} \supseteq \lambda\{f \geq t\} + (1 - \lambda)\{g \geq t\}$, apply Problem 1, integrate in t and use the AM–GM inequality.*)
 3. Deduce Theorem 1 in $\mathbb{R}^n$ from the one-dimensional case by induction on the dimension.
 4. Let $A, B \subseteq \mathbb{R}^n$ be nonempty compact and $\lambda \in (0, 1)$.
 - a) Prove $|\lambda A + (1 - \lambda)B| \geq |A|^\lambda |B|^{1-\lambda}$.

- b) Deduce Theorem 2. (*Hint: apply (a) to $|A|^{-1/n}A$ and $|B|^{-1/n}B$ with $\lambda = |A|^{1/n}/(|A|^{1/n} + |B|^{1/n})$.*)

5. (How sharp is Theorem 1?)

- a) Show that the conclusion cannot be strengthened to $\int h \geq \lambda \int f + (1 - \lambda) \int g$.
- b) Show that it can be, provided $\sup f = \sup g$.
- c) Let $\alpha \in (0, 1)$, $\alpha \neq \lambda$. Show that the hypothesis $h(\lambda x + (1 - \lambda)y) \geq f(x)^\alpha g(y)^{1-\alpha}$ does not imply $\int h \geq (\int f)^\alpha (\int g)^{1-\alpha}$: the exponents must match the way the arguments are mixed. (*Hint: the indicators of $[0, M]$ and $[0, 1]$ with M slightly larger than 1.*)